

INFORMATION VISUALISATION

COMPSCI5099 · Exam Cheatsheet · University of Glasgow

Exam: Thursday 07 May 2026 · 14:00 · 90 min · 4 × 15-mark questions · Closed Book

① KEY DEFINITIONS

Term	Definition
InfoVis	Computer-supported, interactive, visual representation of abstract data to amplify cognition.
SciVis	Visualisation of physically-based data (3D volumes, MRI). Data has inherent spatial meaning.
InfoVis vs SciVis	InfoVis = abstract data, designer chooses layout. SciVis = data already has natural space.
Attribute	A column in your dataset — one measurable property of each item (e.g. age, country, weight).
Mark	Geometric primitive representing a data item: point (0D), line (1D), area (2D).
Channel	Visual property encoding data values: position, colour, size, shape, orientation, texture.
Data-ink Ratio	Ink used for data ÷ total ink. Maximise it. Remove all non-data ink (chartjunk).
Lie Factor	Effect size in graphic ÷ effect size in data. Should be ≈ 1.0 . Truncated y-axis = lie factor > 1 .
Chartjunk	Visual elements that carry no data: 3D effects, heavy grids, decorative images.
Expressiveness	A vis encodes ALL and ONLY the data. No more, no less.
Effectiveness	Of all correct encodings, choose the most effective for the data type and task.
Brushing & Linking	Select items in one view → they highlight in all linked views simultaneously.
Focus + Context	Detail shown in focus area while surrounding context remains visible (e.g. fisheye).
Small Multiples	Repeated similar charts each showing a different subset. Eyes beat memory.
Dynamic Queries	Sliders/toggles that filter data with immediate real-time visual feedback.

② DATA TYPES — What kind of data is it?

Munzner's Attribute Types

Type	Operations	Example
Nominal	$= \neq$	Country, Blood Type, Ward
Ordinal	$= \neq < >$	Low/Med/High, Grades, T-shirt size
Quantitative	$= \neq < > + -$	Age, Weight, Salary, Temperature

■ **Nominal:** No order — just labels. **Ordinal:** Order exists but gaps not equal. **Quantitative:** Real numbers, arithmetic works.

Shneiderman's 7 Data Types — "1-2-3-T-M-Tr-N"

Code	Type	Example	Vis
1D	Linear	Text, lists	Timeline
2D	Planar	Maps, plans	Map/scatter
3D	Volumetric	Molecules, MRI	3D render
T	Temporal	Stock prices	Line chart
M	Multi-dim	DB records	Parallel coords
Tr	Tree	File system	Treemap
N	Network	Social graph	Node-link

③ BERTIN'S VISUAL VARIABLES — Which variable for which data?

Variable	Nominal	Ordinal	Quantitative	Selective?	Associative?	Real-World Use
Position (X,Y)	✓ Good	✓ Good	✓ BEST	✓ Yes	✓ Yes	Scatter plot axes, bar height
Size	■ Marginal	✓ Good	✓ Good	✓ Yes	✗ No	Bubble size for population
Shape	✓ Good	✗ Poor	✗ Poor	✗ NO ★	✓ Yes	Different markers per category
Hue (Colour)	✓ Good	✗ Poor	✗ Poor	✓ Yes	✓ Yes	Country colour on a map
Value (Lightness)	✗ Poor	✓ Good	■ Marginal	✓ Yes	✗ No	Risk level: light→dark
Orientation	■ Marginal	✗ Poor	✗ Poor	✓ Yes	✓ Yes	Wind direction arrows
Texture	✓ Good	■ Marginal	✗ Poor	✓ Yes	✓ Yes	Fine vs coarse hatching

★ **Shape is the ONLY non-selective variable.** Position & Size are the ONLY quantitative variables. Use Hue for nominal, Value for ordinal.

④ SHNEIDERMAN — Mantra + 7 Tasks + 7 Data Types

MANTRA (Memorise This Verbally) "Overview first, zoom and filter, then details-on-demand" — Ben Shneiderman (1996)

Task	What it means	Apply when...
① Overview	Gain a view of ALL data at once	User opens the system — show everything first
② Zoom	Magnify items of interest	User sees a cluster/pattern and wants more detail
③ Filter	Remove uninteresting items	User wants to focus on a subset (e.g. year 2024)
④ Details-on-demand	Click/hover for full record	User found an item and needs all its attributes
⑤ Relate	View relationships between items	User wants to see connections (e.g. network links)
⑥ History	Keep undo/replay states	User wants to backtrack an exploration step
⑦ Extract	Export a sub-collection	User wants to save filtered results as CSV

Mnemonic: **OZF-DRHE** — Overview, Zoom, Filter, Details, Relate, History, Extract

⑤ THE VISUALISATION PIPELINE

Stage	What happens	Example (patient data)	Key decision
1. Raw Data	Original source files	CSV from 100 hospitals	None
2. Data Transform ★	Clean, filter, aggregate, derive	Remove dupes, fill missing, group by month	What to show?
3. Visual Mapping ★★	Map attributes to marks & channels	Date→x-axis, Count→bar height, Hospital→hue	Most critical stage
4. View Transform	Zoom, pan, navigate	User zooms into Q3 2024	What's in view?
5. Rendering	Draw pixels on screen	Browser draws coloured bars	Performance

★ **Visual Mapping is the most critical stage** — this is where mark/channel decisions are made. A wrong choice here (e.g. hue for ordinal data) cannot be fixed later.

⑥ VISUAL PERCEPTION — Pre-attentive Processing & Stevens' Law

Pre-attentive Processing

Features detected in <200ms, in parallel, before conscious attention — the "pop-out" effect.

Pre-attentive Feature	Example
Colour (hue & value)	Red dot among blue dots
Size (length, area)	Large circle among small ones
Orientation	Diagonal line among verticals
Shape (basic)	Circle among squares
Motion / flicker	Moving item among still ones
Spatial position	Outlier far from a cluster
Enclosure	Items inside a box

! **Conjunction search is NOT pre-attentive.** Finding a "red circle" among red squares AND blue circles requires serial scanning — you must check each item one by one (Treisman 1985).

Gestalt Principles (Grouping)

Principle	Meaning	Use in Vis
Proximity	Close items = grouped	Small multiples spacing
Similarity	Alike items = grouped	Colour coding categories
Continuity	Eye follows smooth paths	Line charts
Connectedness	Linked items = grouped	Network edges
Enclosure	Items in boundary = grouped	Panel borders, bounding box
Common fate	Moving together = grouped	Animation

Stevens' Power Law

$$P = k \times S^n \text{ (Perceived magnitude = constant} \times \text{physical stimulus}^{\text{exponent}})$$

Visual Channel	Exponent (n)	Accuracy
Position / Length	≈ 1.0	✓ Accurate
Area	≈ 0.7	■ Underestimated ~30%
Volume	≈ 0.6	✗ Severely wrong
Brightness	≈ 0.5	✗ Very inaccurate

Cleveland & McGill — Perceptual Accuracy Ranking

Most → Least Accurate for quantitative judgements:

Rank	Channel	Chart that uses it
① BEST	Position on common scale	Bar chart, Scatter plot
②	Position on unaligned scales	Grouped charts
③	Length / Height	Bar chart (the bar itself)
④	Angle / Slope	Pie chart ← that's why pies are bad
⑤	Area	Bubble chart, Treemap
⑥	Volume	3D charts ← avoid
⑦ WORST	Colour luminance / saturation	Heatmap



Why bars beat pies: Bars use channel ① (position). Pies use channel ④ (angle). Position perception is far more accurate. Always use this reasoning in exam answers.

Weber's Law (JND)

$\Delta S / S = k$ — the Just Noticeable Difference is proportional to the stimulus. Small differences are easy to detect at low magnitudes; hard at large magnitudes. This is why position beats all other channels — its JND is smallest.

⑦ COLOUR THEORY — Sequential / Diverging / Categorical

Colour Map Types

Type	Structure	Use for	Example
Sequential	One hue, light→dark	Ordered, one-direction data	Rainfall, age, density
Diverging	Two hues from neutral midpoint	Data with meaningful zero	Temp anomaly: blue-white-red
Categorical	Distinct hues, similar lightness	Nominal categories (≤8)	Countries, product types

✗ **Rainbow map = BAD.** Not perceptually uniform → false boundaries where hue changes abruptly. Creates misleading patterns. Use: Viridis, Inferno, Magma instead.

Colour Blindness & Rules

~8% males, ~0.5% females have colour vision deficiency. Protanopia/Deuteranopia = cannot distinguish red from green (most common).

Rule	Why
Never red+green as only cue	8% of males can't distinguish them
Use redundant encoding	Colour + shape, or colour + texture
Use colour-blind-safe palettes	ColorBrewer, Viridis family
Limit categorical hues to ≤8	More = impossible to distinguish

Colour Spaces

Space	Perceptually Uniform?	Use
RGB	✗ No	Screens — not for design
HSL / HSV	✗ No	Design tools — intuitive control
CIELAB	✓ Yes	Scientific vis, colour map design

⑧ CHART TYPES — When to use what

Chart	Data Type	✓ Use when...	✗ Avoid when...	Channel used
Bar chart	Categorical × Quantitative	Comparing values across categories	Showing trends over time	Position (rank ①)
Scatter plot	Quant × Quant	Showing correlation, clusters, outliers	Data is categorical; very large N (overplotting)	Position (rank ①)
Line chart	Temporal × Quant	Trends over time; ordered x-axis	Categories on x-axis (implies false continuity)	Position + direction
Pie chart	Parts of whole (≤5 cats)	Only rough proportions needed	Precise comparison; more than 5 slices	Angle (rank ④) — poor
Histogram	Distribution of 1 variable	Seeing shape of distribution	Comparing multiple distributions	Position + length
Box plot	Distribution summary	Comparing distributions across groups	When bimodality matters (hides shape)	Position + length
Heatmap	Matrix / 2 categorical vars	Large matrices; pattern detection	Precise values needed; poor colour ordering	Colour value (rank ⑦)
Bubble chart	Quant × Quant × Quant	Showing 3rd variable roughly	Precise comparison (area underestimated, $n=0.7$)	Position + area (rank ⑤)

⑨ MULTIDIMENSIONAL VISUALISATION TECHNIQUES

Technique	Max Dims	How it works	✓ Strengths	✗ Weaknesses	Key interaction
Parallel Coords	10–20	Each dim = vertical axis. Each item = polyline connecting values.	Shows all dims at once. Correlations visible between adjacent axes.	Only adjacent-axis correlations. Cluttered with many items.	Brushing on axis
SPLOM	8–12	Grid of ALL pairwise scatter plots. $n \times (n-1)$ total plots.	All pairwise relations visible simultaneously.	Scales poorly (n^2 plots). Small individual plots.	Brushing + linking
Star / Radar	5–10	Radial axes from centre. Each item = polygon.	Good for comparing individual profiles.	Area distorted by axis ordering.	Minimal
Heatmap	50+	Matrix; rows × cols; colour = value.	Compact. Handles very large matrices.	Needs meaningful ordering. Colour limits precision.	Reorder rows/cols
Chernoff Faces	10–15	Data dims mapped to facial features.	Exploits face-recognition ability.	Arbitrary mapping. Some features more salient.	None
Dim. Reduction (PCA/t-SNE/UMAP)	100+	Project high-D to 2D scatter plot.	Overview of very high-D data. Clusters emerge.	Loses individual dimension interpretability.	Select clusters

★ **Parallel lines between 2 axes = positive correlation. Crossing lines (X-shape) = negative correlation.** SPLOM formula: n dimensions → $n \times (n-1)$ scatter plots. For 9 dims = 72 plots.

Algorithm	Type	Preserves	Best for
PCA	Linear	Global variance	When data has linear structure
t-SNE	Non-linear	Local neighbourhood only	Cluster visualisation. Does NOT preserve inter-cluster distances well.
UMAP	Non-linear	Local AND global structure	Best of both — recommended for large datasets
MDS / Spring Model	Distance-based	Pairwise distances	When similarity metric drives layout

■ **Exam trap:** If task needs BOTH fine local detail AND accurate global cluster relationships → t-SNE alone is NOT enough (poor global). UMAP handles both. Spring model/MDS can distort local structure.

⑩ LARKIN & SIMON (1987) — Why a Diagram is (Sometimes) Worth Ten Thousand Words

Concept	Definition	Example
Informational Equivalence	Two representations from which the same information can be derived.	A map and written directions to the same location — same info.
Computational Equivalence	Two representations requiring the same cognitive effort to use.	Map vs directions are NOT computationally equivalent — map is easier to use.

3 Advantages of Diagrams

When Diagrams are NOT better

Advantage	What it means	Situation	Use text instead
Locality	Related info grouped spatially → no search needed	Symbolic / algebraic manipulation	Solving equations, formal proofs
Minimise labelling	Location itself carries info → no need for explicit text labels	Layout doesn't map to problem	Forced spatial layout misleads
Perceptual inference	Some inferences made by visual system directly (e.g. lines cross)	Poor diagram design	A bad diagram is worse than text
		Sequential reasoning needed	Step-by-step logical argument

★ **KEY EXAM POINT:** Two representations can be informationally equivalent (contain identical info) but NOT computationally equivalent (differ hugely in cognitive effort to use). Visualisation amplifies cognition by reducing computational effort — not by adding information.

■ NETWORK & GRAPH VISUALISATION

	Node-Link Diagram	Adjacency Matrix
Structure	Nodes (circles) + edges (lines)	Grid: rows/cols = nodes, cells = edges
Best for	Sparse graphs. Path-tracing tasks. Topology exploration.	Dense graphs. Connection-checking. Large N.
Strengths	Clusters emerge. Intuitive. Force-directed layout reveals structure.	No edge crossings. Good for dense networks. Exact lookup.
Weaknesses	Becomes "hairball" for dense graphs. Overplotting.	Hard to trace paths. Requires meaningful ordering.
Task: "Is A connected to B?"	Hard — visual path-tracing	Easy — find row A, col B. Look at cell.
Task: "Find communities"	Clusters appear visually	Dense diagonal blocks after reordering
Task: "Find influencers"	High-degree nodes = central hubs	Dense rows/columns

Tree Visualisation Techniques

Technique	Best For
Node-link tree	Small-medium hierarchies; shows structure clearly
Treemap	Large hierarchies with quantitative attribute (e.g. file sizes)
Sunburst	Radial treemap; hierarchical navigation
Icicle plot	Wide hierarchies with clear levels

Layout Algorithms

Algorithm	How it works	Weakness
Force-directed	Nodes repel (charge); edges attract (spring). Iterative.	Slow for large graphs; non-deterministic
Sugiyama (layered)	Nodes in layers; minimise crossings	Only for DAGs (directed acyclic)
Circular	Nodes on a circle; edges inside	Hard to trace paths
Matrix	Adjacency matrix with colour	Can't trace paths visually

■ TEMPORAL DATA VISUALISATION

Technique	Use when...	Key strength
Line chart	Single time series; showing trends	Position on x=time axis. Most common temporal vis.
Stacked area chart	Multiple series showing composition over time	Cumulative totals visible
Horizon chart	Many time series in compact space	Colour-coded bands; space-efficient
Sparklines (Tufte)	Embedding trends in text/tables	Tiny; no axes needed; inline
Small multiples	Comparing patterns across categories over time	Direct comparison; no memory needed
Calendar heatmap	Cyclic patterns (day-of-week, seasonal effects)	Day × week grid; colour = value
Gantt chart	Project scheduling; intervals; durations	Shows overlap and sequence of events
Streamgraph / ThemeRiver	Thematic changes over time	Organic stacked areas; centred

	Animation	Small Multiples
Change detection	✗ Hard — change blindness!	✓ Easy — side by side
Comparison	✗ Relies on memory	✓ Direct visual comparison
Engagement	✓ Dynamic, engaging	■ Static
Best for	Presentation / storytelling	Analysis / comparison

! **Animation vs Small Multiples:** Animation suffers from change blindness — users cannot compare "before" and "after" without relying on memory. Small multiples allow direct side-by-side comparison. For ANALYSIS tasks, small multiples almost always win.

■ DESIGN & EVALUATION — Munzner's Nested Model + Methods

Level	Question	Design Threat	Evaluation Method
1. Domain Characterisation	What do users actually need?	Wrong problem — solving the right thing wrongly	Field studies; interviews with domain experts
2. Data/Task Abstraction	How to map domain → data types & tasks?	Wrong abstraction — right problem, wrong model	Observe user workflows; card sorting

Level	Question	Design Threat	Evaluation Method
3. Vis Encoding / Interaction	Which marks, channels, interactions?	Ineffective encoding — correct model, bad vis	Controlled lab experiments; heuristic eval
4. Algorithm / Implementation	How to compute efficiently?	Slow algorithm — correct encoding, bad performance	Benchmarking; profiling

Evaluation Methods

Method	Use when...	Weakness
Controlled Experiment (Quant)	Need to PROVE A is faster/better than B statistically	Lab ≠ real world; limited task scope
Think-aloud / Interview (Qual)	Need to understand WHY users struggle	Subjective; hard to generalise; small N
Heuristic Evaluation	Need quick expert review of design	Expert views ≠ real user behaviour
Case Study / Deployment	Need ecological validity with real users	Hard to control; can't generalise easily

Evaluation Metrics

Metric	Measures
Task completion time	Efficiency
Error rate	Accuracy
Insight count	Discovery effectiveness
Likert scale (1–7)	Subjective satisfaction
Learning curve	Learnability over sessions

Tufte's Principles

Principle	Rule
Data-ink ratio	Maximise. Remove all non-data ink.
Chartjunk	Delete: 3D effects, heavy grids, decoration
Lie factor	Should be ≈ 1.0. Truncated axis = lie.
Small multiples	Better than animation for comparison

■ WHEN TO APPLY WHAT — Quick Decision Guide

Use this when you see a scenario in the exam

IF the question involves...	APPLY / USE...	KEY REFERENCE
Choosing which visual property (colour/size/shape) to use for data	Bertin's visual variables → match data type to correct variable	Bertin (1967)
Why a chart is good or bad at showing quantities	Stevens' law + Cleveland & McGill ranking → state the exponent and rank	Stevens + Cleveland (1984)
Designing interaction for a complex dataset	Shneiderman's mantra + 7 tasks → apply overview→filter→details	Shneiderman (1996)
Why a diagram helps vs text/prose	Larkin & Simon → informational vs computational equivalence; 3 advantages	Larkin & Simon (1987)
User selects/queries data interactively	Heer's generalised selection → declarative query + relaxation. Compare with dynamic queries	Heer et al.
Data with many attributes (5+ dimensions)	Parallel coordinates or SPLOM → describe interaction (brushing/reordering)	Munzner (2015)
Visualising a network or graph	Node-link vs matrix → decide based on density + task type	Graph vis theory
Data changes over time	Line chart / calendar heatmap / small multiples. Animation vs small multiples argument	Shneiderman + Tufte
Evaluating whether a vis system works	Munzner's nested model → 4 levels + threats. Quant vs qual evaluation	Munzner (2015)
Choosing a colour map	Sequential (ordered), Diverging (meaningful zero), Categorical (nominal). Avoid rainbow.	Colour theory
Any design critique question	Tufte: data-ink ratio, chartjunk, lie factor. Munzner: no unjustified 3D, eyes beat memory	Tufte + Munzner
What type of data is it?	Nominal / Ordinal / Quantitative (Munzner). Shneiderman's 7 data types for structure.	Both frameworks
SQL is error-prone / hard to learn	Dynamic queries → sliders/dropdowns, immediate feedback, no syntax errors possible	Shneiderman
Showing many categories over time	Small multiples — one panel per category. Not animation (change blindness).	Tufte

■ KEY PEOPLE & PAPERS — Who said what

Person	Year	Key Contribution	Magic Phrase
Jacques Bertin	1967	Sémiologie Graphique — 7 retinal variables + position. Properties: associative, selective, ordered, quantitative.	"Position is the most powerful visual variable"
Ben Shneiderman	1996	"The Eyes Have It" — Visual Information-Seeking Mantra. Task by Data Type Taxonomy (7×7).	"Overview first, zoom and filter, then details-on-demand"
Edward Tufte	1983+	Data-ink ratio, chartjunk, lie factor, small multiples.	"Above all else, show the data"
Tamara Munzner	2015	Nested model (4 levels). Marks and channels. Effectiveness ranking.	"Expressiveness + Effectiveness"
Cleveland & McGill	1984	Empirical perceptual accuracy hierarchy. Position > Length > Angle > Area > Volume > Colour.	"Position on a common scale is most accurate"
Jeffrey Heer	~2008	Generalised selection via interactive query relaxation. Bottom-up query building from data examples.	"Selection as a declarative query"
Larkin & Simon	1987	"Why a Diagram is (Sometimes) Worth Ten Thousand Words." Informational vs computational equivalence.	"Computational efficiency, not informational content"
Anne Treisman	1985	Feature Integration Theory. Pre-attentive processing. Conjunction search = serial, not parallel.	"Pop-out effect" — single features processed in parallel
Colin Ware	2004+	Information Visualization: Perception for Design. Vision science applied to vis.	Perception-first design approach
Stevens	1957	Power Law: $P = k \times S^n$. Different channels have different exponents. Area underestimated ($n \approx 0.7$).	"Perceived magnitude follows a power function"

■ COMMON DESIGN PITFALLS — Spot these in critique questions

Pitfall	Why it's wrong	Fix it by...
Truncated y-axis (bars not starting at 0)	Lie factor >> 1. Exaggerates differences visually.	Start y-axis at 0 for bar charts always.

Pitfall	Why it's wrong	Fix it by...
Pie chart for comparison	Angle perception = rank ④ in accuracy. Hard to compare non-adjacent slices.	Use bar chart (position = rank ①).
Rainbow colour map	Not perceptually uniform. Creates false visual boundaries. Fails for colour-blind users.	Use Viridis, Inferno, Magma.
Unjustified 3D charts	Occlusion + perspective distortion. Volume exponent $n \approx 0.6$ = severe underestimation.	Use 2D unless data is inherently 3D.
Dual y-axes on same chart	Can create misleading apparent correlations between unrelated variables.	Use two separate charts side by side.
Overplotting in scatter plots	Dense points overlap — can't see distribution or clusters.	Use transparency (alpha), jitter, or aggregation.
Using hue for ordinal data	Hue is not ordered — Bertin says it's poor for ordinal. Implies no ranking.	Use colour value (lightness) for ordered data.
Using size for nominal data	Size implies order. Bigger = more. Nominal data has no order.	Use hue or shape for nominal categories.
Area encoding for precise comparison	$n \approx 0.7 \rightarrow$ systematic underestimation. Users think big bubbles are smaller than they are.	Use length/position instead. Label bubbles explicitly.
Animation for analytical comparison	Change blindness. Users can't compare before/after without memory.	Use small multiples for analysis tasks.

For every 15-mark question: **(1)** Classify the data type **(2)** Choose marks & channels **(3)** Apply Shneiderman's mantra to interaction **(4)** Reference the relevant person/paper **(5)** Justify your choice said: think "how would I USE this?" not "what are all the details of this?"